

```
-- =====
-- IPL ANALYTICS USING POSTGRESQL
-- Dataset: IPL ball-by-ball data
-- Table: ipl_data
-- =====

-- QUESTION 1
-- How has the scoring pattern of IPL changed across seasons?
-- This analysis compares the average innings score and run rate
-- for every season. It helps us understand whether batting has
-- become more aggressive over the years.

WITH innings_data AS (
    SELECT
        season,
        match_id,
        innings,
        SUM(runs_total) AS innings_runs,
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls
    FROM ipl_data
    GROUP BY season, match_id, innings
)

SELECT
    season,
    COUNT(DISTINCT match_id) AS matches,
    ROUND(AVG(innings_runs), 2) AS average_innings_score,
    ROUND(
        SUM(innings_runs) * 6.0 /
        NULLIF(SUM(legal_balls), 0),
        2
    ) AS run_rate
FROM innings_data
GROUP BY season
ORDER BY season;
```

	season text	matches bigint	average_innings_score numeric	run_rate numeric
1	2007/08	58	154.63	8.31
2	2009	57	140.97	7.49
3	2009/10	60	154.78	8.13
4	2011	73	145.89	7.73
5	2012	74	151.71	7.83
6	2013	76	144.88	7.69
7	2014	60	155.17	8.20
8	2015	59	154.23	8.37
9	2016	60	157.18	8.31
10	2017	59	156.55	8.41
11	2018	60	165.84	8.65

```
-- QUESTION 2
-- Which teams are the most consistent at scoring runs?
-- Instead of looking only at total runs, this analysis looks at
-- average score, highest score, lowest score and score variation.
-- A lower variation means the team scores more consistently.
```

```
WITH team_innings AS (
    SELECT
        batting_team,
        match_id,
        innings,
        SUM(runs_total) AS total_runs
    FROM ipl_data
    GROUP BY batting_team, match_id, innings
)
```

```
SELECT
    batting_team,
    COUNT(*) AS innings_played,
    ROUND(AVG(total_runs), 2) AS average_score,
    ROUND(STDDEV(total_runs), 2) AS score_variation,
    MIN(total_runs) AS lowest_score,
    MAX(total_runs) AS highest_score
FROM team_innings
GROUP BY batting_team
HAVING COUNT(*) >= 20
ORDER BY average_score DESC;
```

	batting_team text	innings_played bigint	average_score numeric	score_variation numeric	lowest_score numeric	highest_score numeric
1	Royal Challengers Bengal...	46	187.83	38.14	77	262
2	Gujarat Titans	77	178.48	32.25	89	233
3	Punjab Kings	87	178.26	37.21	98	265
4	Lucknow Super Giants	73	174.71	38.67	1	257
5	Chennai Super Kings	266	163.91	32.40	9	246
6	Delhi Capitals	123	162.27	42.75	3	264
7	Sunrisers Hyderabad	214	161.87	42.86	2	287
8	Mumbai Indians	296	160.95	37.34	5	247
9	Rajasthan Royals	254	159.20	39.30	6	243
10	Gujarat Lions	31	156.84	37.17	6	208
11	Royal Challengers Bangal...	243	155.11	38.74	2	263
12	Mumbai Indians	222	154.88	38.18	2	278

```
-- QUESTION 3
-- Which teams are the most successful when chasing a target?
-- This compares the first innings score with the second innings
-- score and calculates the percentage of successful chases.
-- It helps identify teams that perform well under chasing pressure.
```

```
WITH innings_scores AS (
    SELECT
        match_id,
        innings,
        batting_team,
```

```

        SUM(runs_total) AS runs
    FROM ipl_data
    GROUP BY match_id, innings, batting_team
),
match_scores AS (
    SELECT
        match_id,

        MAX(
            CASE
                WHEN innings = 1 THEN runs
            END
        ) AS first_innings_score,

        MAX(
            CASE
                WHEN innings = 2 THEN batting_team
            END
        ) AS chasing_team,

        MAX(
            CASE
                WHEN innings = 2 THEN runs
            END
        ) AS chasing_score

    FROM innings_scores
    GROUP BY match_id
)

SELECT
    chasing_team,
    COUNT(*) AS chases,

    COUNT(*) FILTER (
        WHERE chasing_score > first_innings_score
    ) AS successful_chases,

    ROUND(
        COUNT(*) FILTER (
            WHERE chasing_score > first_innings_score
        ) * 100.0 / COUNT(*),
        2
    ) AS chase_success_rate

FROM match_scores
WHERE chasing_team IS NOT NULL
GROUP BY chasing_team
HAVING COUNT(*) >= 10
ORDER BY chase_success_rate DESC;

```

	chasing_team text	chases bigint	successful_chases bigint	chase_success_rate numeric
1	Gujarat Lions	16	12	75.00
2	Royal Challengers Bengal...	22	16	72.73
3	Gujarat Titans	43	27	62.79
4	Punjab Kings	40	24	60.00
5	Chennai Super Kings	121	72	59.50
6	Mumbai Indians	141	78	55.32
7	Kolkata Knight Riders	145	80	55.17
8	Rajasthan Royals	133	73	54.89
9	Delhi Capitals	67	35	52.24
10	Royal Challengers Bangal...	119	60	50.42
11	Sunrisers Hyderabad	98	48	48.98

```
-- QUESTION 4
-- Does winning the toss actually increase the chance of winning
-- the match?
-- This analysis compares the toss winner with the actual match
-- winner. It tells us whether the toss provides a meaningful
-- historical advantage.
```

```
WITH match_results AS (
    SELECT DISTINCT
        match_id,
        toss_winner,
        match_won_by
    FROM ipl_data

    WHERE toss_winner IS NOT NULL
        AND match_won_by IS NOT NULL
)

SELECT
    COUNT(*) AS total_matches,

    COUNT(*) FILTER (
        WHERE toss_winner = match_won_by
    ) AS toss_winner_wins,

    ROUND(
        COUNT(*) FILTER (
            WHERE toss_winner = match_won_by
        ) * 100.0 / COUNT(*),
        2
    ) AS toss_win_percentage

FROM match_results;
```

	total_matches bigint	toss_winner_wins bigint	toss_win_percentage numeric
1	1243	628	50.52

```
-- QUESTION 5
-- Which teams score most aggressively during the Powerplay?
-- The first six overs can strongly influence an innings.
-- This query calculates the average Powerplay score and run rate
-- for each team, allowing us to compare their early-innings
-- attacking ability.
```

```
WITH powerplay AS (
    SELECT
        batting_team,
        match_id,

        SUM(runs_total) AS runs,

        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls

    FROM ipl_data

    WHERE over BETWEEN 0 AND 5

    GROUP BY batting_team, match_id
)

SELECT
    batting_team,

    COUNT(*) AS innings,

    ROUND(
        AVG(runs),
        2
    ) AS average_powerplay_runs,

    ROUND(
        SUM(runs) * 6.0 /
        NULLIF(SUM(legal_balls), 0),
        2
    ) AS powerplay_run_rate

FROM powerplay

GROUP BY batting_team

HAVING COUNT(*) >= 20

ORDER BY powerplay_run_rate DESC;
```

	batting_team text	innings bigint	average_powerplay_runs numeric	powerplay_run_rate numeric
1	Royal Challengers Bengal...	46	59.96	9.99
2	Punjab Kings	87	54.83	9.14
3	Gujarat Titans	77	51.92	8.65
4	Delhi Capitals	119	51.98	8.63
5	Lucknow Super Giants	72	51.79	8.62
6	Gujarat Lions	30	51.97	8.61
7	Sunrisers Hyderabad	210	50.93	8.48
8	Rajasthan Royals	250	49.08	8.18
9	Kolkata Knight Riders	278	48.16	8.05
10	Mumbai Indians	291	47.90	7.97
11	Chennai Super Kings	265	47.40	7.90

-- QUESTION 6
 -- Which teams accelerate the most from the Powerplay to the
 -- Death Overs?
 -- This compares the run rate during overs 0-5 with the run rate
 -- during overs 16-19. A high acceleration value indicates that
 -- a team becomes significantly more aggressive at the end
 -- of its innings.

```
WITH phase_scores AS (
  SELECT
    batting_team,
    match_id,

    SUM(
      CASE
        WHEN over BETWEEN 0 AND 5
        THEN runs_total
        ELSE 0
      END
    ) AS powerplay_runs,

    SUM(
      CASE
        WHEN over BETWEEN 16 AND 19
        THEN runs_total
        ELSE 0
      END
    ) AS death_runs,

    COUNT(*) FILTER (
      WHERE over BETWEEN 0 AND 5
      AND valid_ball = 1
    ) AS powerplay_balls,

    COUNT(*) FILTER (
      WHERE over BETWEEN 16 AND 19
      AND valid_ball = 1
    ) AS death_balls
)
```

```

        ) AS death_balls

FROM ipl_data

GROUP BY batting_team, match_id
)

SELECT
    batting_team,

    ROUND(
        SUM(powerplay_runs) * 6.0 /
        NULLIF(SUM(powerplay_balls), 0),
        2
    ) AS powerplay_run_rate,

    ROUND(
        SUM(death_runs) * 6.0 /
        NULLIF(SUM(death_balls), 0),
        2
    ) AS death_run_rate,

    ROUND(
        (
            SUM(death_runs) * 6.0 /
            NULLIF(SUM(death_balls), 0)
        )
        -
        (
            SUM(powerplay_runs) * 6.0 /
            NULLIF(SUM(powerplay_balls), 0)
        ),
        2
    ) AS acceleration

FROM phase_scores

GROUP BY batting_team

ORDER BY acceleration DESC;

```

```

-- QUESTION 7
-- Which batsmen have the best strike rate while maintaining
-- a meaningful sample size?
-- Ranking only by strike rate can be misleading because a player
-- with very few balls may appear at the top. Here we consider
-- only batsmen who have faced at least 500 legal balls.

```

```

WITH batting AS (
    SELECT
        batter,

        SUM(runs_batter) AS runs,

        COUNT(*) FILTER (
            WHERE valid_ball = 1

```

```

        ) AS balls

    FROM ipl_data

    GROUP BY batter
)

SELECT
    batter,
    runs,
    balls,

    ROUND(
        runs * 100.0 /
        NULLIF(balls, 0),
        2
    ) AS strike_rate

FROM batting

WHERE balls >= 500

ORDER BY strike_rate DESC

LIMIT 20;

```

	batter text	runs numeric	balls bigint	strike_rate numeric
1	TH David	1151	642	179.28
2	AD Russell	2655	1508	176.06
3	PD Salt	1258	715	175.94
4	Abhishek Shar...	2379	1383	172.02
5	TM Head	1556	907	171.55
6	H Klaasen	2104	1253	167.92
7	SP Narine	1820	1099	165.61
8	RM Patidar	1612	978	164.83
9	N Pooran	2527	1538	164.30
10	Shashank Singh	905	560	161.61
11	P Simran Singh	1865	1184	157.52

```

-- QUESTION 8
-- Which bowlers create the most dot-ball pressure?
-- Wickets are not the only measure of bowling quality.
-- Dot balls create pressure and can force batsmen to take
-- risky shots. This query compares dot-ball percentage among
-- bowlers with a reasonable number of deliveries.

```

```

SELECT
    bowler,

```

```

COUNT(*) FILTER (
    WHERE valid_ball = 1
) AS legal_balls,

COUNT(*) FILTER (
    WHERE valid_ball = 1
    AND runs_total = 0
) AS dot_balls,

ROUND(
    COUNT(*) FILTER (
        WHERE valid_ball = 1
        AND runs_total = 0
    ) * 100.0 /

    NULLIF(
        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ),
        0
    ),

    2
) AS dot_ball_percentage

FROM ipl_data

GROUP BY bowler

HAVING COUNT(*) FILTER (
    WHERE valid_ball = 1
) >= 500

ORDER BY dot_ball_percentage DESC;

```

Data Output Messages Notifications				
	bowler text	legal_balls bigint	dot_balls bigint	dot_ball_percentage numeric
1	DW Steyn	2182	1019	46.70
2	S Sreesanth	880	399	45.34
3	DP Nannes	646	285	44.12
4	DE Bollinger	576	254	44.10
5	RJ Harris	832	364	43.75
6	MM Patel	1355	590	43.54
7	M Morkel	1629	708	43.46
8	MG Johnson	1235	535	43.32
9	NM Coulter-Nile	857	368	42.94
10	JC Archer	1578	677	42.90
11	B Lee	875	374	42.74

```
-- QUESTION 9
-- Which venues are the most batting-friendly?
-- Different stadiums can produce very different scoring patterns.
-- This query calculates average match runs at each venue so that
-- we can identify venues where high-scoring matches are more common.
```

```
WITH venue_matches AS (
  SELECT
    venue,
    match_id,
    SUM(runs_total) AS total_runs

  FROM ipl_data

  GROUP BY venue, match_id
)

SELECT
  venue,

  COUNT(*) AS matches,

  ROUND(
    AVG(total_runs),
    2
  ) AS average_match_runs,

  ROUND(
    STDDEV(total_runs),
    2
  ) AS score_variation
```

```

FROM venue_matches

GROUP BY venue

HAVING COUNT(*) >= 10

ORDER BY average_match_runs DESC;

```

	venue text	matches bigint	average_match_runs numeric	score_variation numeric
1	Arun Jaitley Stadium, Delhi	30	375.13	74.31
2	Himachal Pradesh Cricket Association Stadium, Dharamsala	10	374.00	96.13
3	Sawai Mansingh Stadium, Jaipur	21	372.81	61.88
4	M Chinnaswamy Stadium, Bengaluru	24	372.08	68.41
5	Rajiv Gandhi International Stadium, Uppal, Hyderabad	26	369.42	83.76
6	Eden Gardens, Kolkata	30	359.43	90.52
7	Wankhede Stadium, Mumbai	59	353.44	64.14
8	Narendra Modi Stadium, Ahmedabad	41	351.90	58.59
9	Brabourne Stadium	10	348.10	42.65
10	Punjab Cricket Association IS Bindra Stadium	10	347.60	26.02
11	Brabourne Stadium, Mumbai	17	342.59	69.61

```

-- QUESTION 10
-- Which teams have the strongest overall batting profile?
-- This combines multiple batting indicators instead of ranking
-- teams using only total runs. It compares runs per match,
-- run rate, boundaries and dot-ball percentage to understand
-- how teams build their innings.

```

```

WITH team_stats AS (
    SELECT
        batting_team,

        SUM(runs_total) AS total_runs,

        COUNT(DISTINCT match_id) AS matches,

        COUNT(*) FILTER (
            WHERE runs_batter = 4
        ) AS fours,

        COUNT(*) FILTER (
            WHERE runs_batter = 6
        ) AS sixes,

        COUNT(*) FILTER (
            WHERE valid_ball = 1
        ) AS legal_balls,

        COUNT(*) FILTER (
            WHERE runs_total = 0
            AND valid_ball = 1
        ) AS dot_balls

    FROM ipl_data

```

```

        GROUP BY batting_team
    )

SELECT
    batting_team,

    matches,

    total_runs,

    ROUND(
        total_runs * 1.0 /
        NULLIF(matches, 0),
        2
    ) AS runs_per_match,

    fours,

    sixes,

    ROUND(
        total_runs * 100.0 /
        NULLIF(legal_balls, 0),
        2
    ) AS run_rate,

    ROUND(
        dot_balls * 100.0 /
        NULLIF(legal_balls, 0),
        2
    ) AS dot_ball_percentage

FROM team_stats

ORDER BY run_rate DESC;

```

	batting_team text	matches bigint	total_runs numeric	runs_per_match numeric	fours bigint	sixes bigint	run_rate numeric	dot_ball_percentage numeric
1	Royal Challengers Bengal...	46	8640	187.83	729	454	168.62	30.72
2	Punjab Kings	87	15509	178.26	1281	782	155.51	34.76
3	Gujarat Titans	77	13743	178.48	1239	519	153.40	30.85
4	Lucknow Super Giants	72	12754	177.14	1004	611	151.15	34.39
5	Delhi Capitals	119	19959	167.72	1798	777	144.92	34.33
6	Sunrisers Hyderabad	210	34640	164.95	2879	1333	142.59	34.09
7	Mumbai Indians	291	47641	163.71	4101	1968	141.51	36.56
8	Gujarat Lions	30	4862	162.07	460	155	141.38	33.59
9	Chennai Super Kings	265	43600	164.53	3615	1735	141.11	34.98
10	Rajasthan Royals	250	40437	161.75	3559	1564	140.89	36.33
11	Kolkata Knight Riders	278	43646	157.00	3826	1711	139.92	36.83